COURSE TITLE : FOUNDATIONS OF COMPUTER HARDWARE

COURSE CODE : 4264

COURSE CATEGORY : A

PERIODS/WEEK : 4

PERIODS/SEMESTER : 60

CREDITS : 4

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Basics of Switching Theory and Logic Design	**15**
2	Combinational Logic Circuits	**15**
3	Sequential Logic Circuits	**15**
4	Basics of Microprocessors	**15**

COURSE GENERAL OUTCOMES:

Sl.	G.O	On completion of this course the student will be able to :
1	**1**	To understand foundations of Switching Theory and Logic Design
2	**1**	To design Combinational Logic Circuits
3	**1**	To design Sequential Logic Circuits
4	**1**	To understand basics of microprocessors

SPECIFIC OUTCOMES:

MODULE – I: Basics of Switching Theory and Logic Design

1.1 To understand foundations of Switching Theory and Logic Design

1.1.1 Describe Basics of Data Representation

1.1.2 Distinguish fundamental number systems- Decimal, Binary, Octal, Hexa-decimal

1.1.3 Practice conversion between various number systems.

1.1.4 Identify various binary codes- numeric & alphanumeric codes

1.1.5 Discuss ASCII code

1.1.6 Describe EBCIDIC code

1.1.7 Explain UNICODE.

1.1.8 Differentiate weighted and unweighted codes.

1.1.9 Familiarise basics of switching theory and Boolean algebra.

1.1.10 Explain basic postulates and theorems of Boolean algebra.

1.1.11 Define De Morgan's theorems.

1.1.12 Practice simplification of logic function using Karnaugh map.

MODULE – II: Combinational Logic Circuits

2.1 To design Combinational Logic Circuits

2.1.1 Familiarise basic logic gates – OR, AND, NOT, XOR, XNOR.

2.1.2 Explain Universal Gates-NOR, NAND.

2.1.3 Design basic logic gates using Universal gates.

2.1.4 Describe basics of combinational logic circuits.

2.1.5 Design Adders (Half Adder, Full Adder, Parallel Adder- Ripple Carry Adder, Carry Look Ahead Adder, Serial Adders).

2.1.6 Design Subtractors(Half-subtractor, Full subtractor).

2.1.7 Design Encoder circuits.

2.1.8 Design decoder circuits

2.1.9 Explain Multiplexers.

2.1.10 Discuss Demultiplexer.

MODULE – III: Sequential Logic Circuits

3.1 To design Sequential Logic Circuits

3.1.1 Explain Basic sequential logic circuits- Flip-flops (SR, JK, D, T, Master-Slave).

3.1.2 Describe SR latch

3.1.3 Explain SR and JK flipflops

3.1.4 Differentiate D and T flipflops.

3.1.5 Describe Master- Slave flipflops

3.1.6 Use State tables, State equations, Excitation tables and state diagrams to design sequential circuits.

3.1.7 Discuss the basics of synchronous counters.

3.1.8 Describe asynchronous counter

3.1.9 Design Synchronous counters(Johnson and Ring counter).

3.1.10 Design Asynchronous counters- Binary, BCD, Up/Down Counters.

MODULE – IV: Basics of Microprocessors

4.1 To understand basics of microprocessors

4.1.1 Discuss various registers.

4.1.2 Explain shift registers.
4.1.3 Design shift registers using flip flops.
4.1.4 Distinguish basic memory elements(RAM, ROM, Flash memory).
4.1.5 Understand basic processor architecture.
4.1.6 Explain pin diagram of Intel 8085 microprocessor
4.1.7 Describe pin diagram of Intel 8086 microprocessor
4.1.8 Explain register organization of 8086 microprocessor.
4.1.9 Explain memory organization of 8086 microprocessor.
4.1.10 Describe basic addressing modes.
4.1.11 List instruction set in 8086.
4.1.12 Describe interrupts in 8086(software and hardware interrupts).
4.1.13 Practice programming with 8086.

CONTENT DETAILS

MODULE – I: Basics of Switching Theory and Logic Design

Data representation & Number Systems - Data representation, Number systems – Decimal, Binary, Octal, Hexa-decimal; Conversion between various number systems;Binary Codes- Alphanumeric codes-ASCII, EBCIDIC,UNICODE; Numeric codes- weighted codes and un-weighted code;

Switching Theory and Logic Design – Boolean algebra – postulates and theorems, De Morgan's theorems, Simplifications of logic functions using Karnaugh map

MODULE – II: Combinational Logic Circuits

Basic logic circuits: Basic logic gates-OR Gate, AND Gate , NOT Gate; Universal Gates-NOR, NAND; other gates – XOR, XNOR;

Combinational Logic circuits: Adders &Subtractors, Serial Adder, Parallel Adder- Carry propagate adder, Carry Look Ahead Adder; Decoders, Encoders; Multiplexers, Demultiplexers;

MODULE – III: Sequential Logic Circuits

Sequential Circuits: Latches & flip-flops- SR, JK, D, T and Master-slave Flip flops; Clocked Sequential Circuits- State Tables State Equations and State Diagrams;

Counters: Design of Synchronous and Asynchronous Counters:- Binary, BCD, Decade and Up/Down Counters;

MODULE – IV: Basics of Microprocessors

Memory Elements:- Registers, Shift Registers, Types of Shift Registers, RAM, ROM, Flash memory;

8085 & 8086:- Basic processor architecture; Intel 8085 and Intel 8086 Microprocessor: Architecture, Pin Diagram, Register Organization, Memory Organization; Programming with 8086 Microprocessor- Instruction Set, Addressing Modes, Programming Examples, 8086 Interrupts- Hardware and Software Interrupts.

TEXT BOOK(S)

1. Morris Mano ,Digital Logic and Computer Design Prentice Hall of India

2. Douglas V.Hall, Microprocessors and Interfacing Tata MCGraw Hill

REFFERENCE

1.Floyd T.L. Digital Fundamentals , Universal Bookstall

2. Brey B.B., The Intel Microprocessors - Architecture, Programming & Interfacing, Prentice Hall.